

Quantum Computing at the Threshold of Utility

A Critical Review of Hardware Platforms, Algorithms, Cryptographic Implications and Commercial Readiness, 2024–2026

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Abstract

Quantum information processing has moved, within the space of roughly two years, from a discipline defined by extrapolation to one defined by measurement. This review consolidates and critically appraises the technical and commercial state of the field as it stood in mid-2026. Four developments frame the analysis: the demonstration of verifiable computational advantage on superconducting hardware through observable-based echo protocols in April 2026; the extension of advantage claims into the simulation of heterogeneous quantum materials in July 2026; the operation of surface-code memories below the fault-tolerance threshold; and the completion of the first tranche of standardised post-quantum cryptographic primitives. Against these advances the review sets an equally important body of negative evidence — systematic benchmark studies in which classically tractable formulations of putative quantum use cases were solved to proven optimality in seconds, and the withdrawal of at least one major financial institution from quantum research after resource estimation placed its target problem beyond any plausible machine. The paper surveys six physical qubit modalities, evaluates them against criteria that extend well past raw qubit count, examines the principal algorithmic families and their known limitations, assesses the software and cloud-access ecosystem, analyses the cryptographic migration timeline in policy and engineering terms, and reviews error-correction progress from repetition codes through quantum low-density parity-check constructions. The central conclusion is that the field has entered a period in which scientific advantage is demonstrable but economic advantage is not, and that the interval between the two will be governed less by qubit counts than by logical error rates, compilation overheads, and the availability of trained personnel.

Keywords: quantum computing; qubit modalities; quantum error correction; fault tolerance; variational algorithms; post-quantum cryptography; quantum advantage; technology readiness.

1. Introduction

1.1 Background and Motivation

Digital computation rests on the manipulation of deterministic binary states. That abstraction, extraordinarily productive for seven decades, becomes inefficient in two specific regimes: the simulation of interacting quantum systems, where the classical memory required grows exponentially with the number of correlated degrees of freedom, and certain structured algebraic problems for which no efficient classical procedure is known. Quantum processors address both regimes by encoding information in the amplitudes of a superposed state and manipulating those amplitudes through interference [1].

The motivation for a fresh review is that the empirical basis for judging the field changed materially during 2024–2026. Where earlier assessments necessarily reasoned from roadmaps, the present one can reason from published error rates, replicated error-correction results, standardised cryptographic primitives, and

— significantly — from documented commercial retreats. The field has acquired a falsification record, and any credible survey must weigh it alongside the announcements.

Three quantitative anchors illustrate how far the hardware has moved. Superconducting processors now report error per layered gate of approximately 2.15×10^{-3} at the hundred-qubit scale [11]. Trapped-ion systems have reported two-qubit gate infidelity below one part in a thousand, with a 98-qubit, all-to-all connected machine quoting 99.921% two-qubit fidelity [8]. Neutral-atom arrays have crossed the thousand-qubit line, with coherence times approaching a minute [9]. None of these figures existed in this combination three years ago.

1.2 Objectives and Research Questions

This review is organised around five questions:

1. Which physical qubit modality, if any, currently offers a credible path to fault-tolerant operation at commercially relevant scale, and on what evidence?
2. Which algorithmic families possess proven asymptotic advantage, which rest on heuristic argument, and where does the boundary between them lie in practice?
3. How severe is the near-term cryptographic exposure created by advances in factoring resource estimation, and what migration posture does the evidence support?
4. Where does demonstrated scientific advantage stand relative to demonstrated economic advantage, and what accounts for the gap?
5. Which open problems most tightly constrain the field’s trajectory over the remainder of the decade?

1.3 Methodology and Evidence Base

Evidence was drawn from three tiers, weighted in the following order. The first tier comprises peer-reviewed publications, principally in *Nature*, *Nature Communications*, *Physical Review* titles and *npj Quantum Information*, together with standards documents issued by the National Institute of Standards and Technology (NIST) and the European Telecommunications Standards Institute (ETSI). The second tier comprises preprints, predominantly from arXiv, which carry the field’s most current results but have not completed review. The third tier comprises vendor technical documentation, corporate research blogs, investor disclosures and specialist trade press.

Third-tier material is used only for claims that are essentially factual and self-reported — pricing, product availability, funding amounts, stated roadmap targets — and is explicitly attributed wherever a performance claim originates there. Vendor performance figures are quoted as reported rather than as independently verified; the distinction matters, because benchmarking conventions differ substantially between manufacturers and few cross-platform results are audited by third parties. Where a claim is contested in the literature, the disagreement is stated rather than resolved.

Two methodological limitations should be recorded at the outset. First, the field’s publication cycle is fast enough that any snapshot degrades within months. Second, negative results are systematically under-reported: failed demonstrations rarely reach publication, which biases the visible record toward optimism. The review compensates where it can by giving explicit weight to the small number of published negative and null results.

1.4 Contributions and Structure

The paper contributes (i) a comparative evaluation of six qubit modalities against a criterion set that extends beyond qubit count to connectivity, gate speed, compilation overhead and cryogenic burden; (ii) an assessment of algorithmic maturity that separates provable from heuristic speedups and accounts for classical dequantisation results; (iii) a consolidated timeline of cryptographic migration mandates across major jurisdictions; and (iv) a structured account of the gap between scientific and commercial advantage.

Section 2 establishes formal preliminaries. Section 3 surveys hardware. Section 4 examines algorithms. Section 5 reviews software frameworks and cloud access. Section 6 treats quantum cryptography and post-quantum migration. Section 7 assesses industry applications and quantum machine learning. Section 8 covers error correction and the systemic obstacles to scaling. Section 9 presents market structure, public investment and corporate roadmaps. Section 10 offers a critical synthesis and research agenda; Section 11 concludes.

2. Theoretical Preliminaries

2.1 State Space and Measurement

The formalism is that of finite-dimensional complex Hilbert spaces. An n -qubit register occupies $\mathcal{H} = \mathbb{C}^{2^n}$, a space whose dimension doubles with each additional qubit. A pure state is written $|\psi\rangle = \sum_i \alpha_i |i\rangle$, where the index runs over the 2^n computational basis strings and the amplitudes $\alpha_i \in \mathbb{C}$ satisfy the normalisation constraint $\sum_i |\alpha_i|^2 = 1$ [1].

Measurement in the computational basis returns outcome i with probability $|\alpha_i|^2$ — the Born rule — and irreversibly projects the register onto $|i\rangle$. This is the structural constraint that distinguishes quantum from probabilistic classical computation: the exponentially large amplitude vector is never directly accessible, and algorithmic design is therefore the art of arranging interference so that the small amount of information a measurement can extract is the information one wants.

Composite systems combine by tensor product, so that two qubits inhabit $\mathcal{H}_A \otimes \mathcal{H}_B \cong \mathbb{C}^4$. States that cannot be written as a product of single-qubit states are entangled, and it is precisely this non-factorisable fraction of the state space that resists efficient classical description. The distinction matters for a common misstatement: exponential state-space dimension is necessary for quantum advantage but nowhere near sufficient, since classical probability distributions over n bits are likewise exponentially parameterised.

2.2 Qubit Representation and the Bloch Sphere

A single qubit generalises the classical bit by admitting superposition, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ subject to $|\alpha|^2 + |\beta|^2 = 1$. Modulo global phase, every pure single-qubit state may be parameterised as $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$ and mapped bijectively to a point on the unit sphere. Mixed states occupy the interior; decoherence is, in this picture, contraction of the Bloch vector toward the origin.

Physical realisations differ in which degree of freedom carries the logical state. Transmon architectures use the two lowest levels of a weakly anharmonic superconducting oscillator formed by a Josephson junction shunted with a large capacitance, a design chosen specifically to suppress charge-noise sensitivity [18]. Trapped-ion machines use long-lived internal electronic or hyperfine states. Neutral-atom platforms increasingly encode information in nuclear spin, which couples weakly to the electromagnetic environment and thereby extends coherence [9].

2.3 Entanglement and Interference

Entangled states such as the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ exhibit correlations with no classical counterpart: measuring one qubit fixes the outcome of the other irrespective of separation, without permitting superluminal signalling. Quantum algorithms exploit interference to add amplitude constructively on solution states while cancelling it elsewhere. Grover's procedure is the canonical illustration — alternating oracle reflection and diffusion rotates the state vector toward the marked subspace in approximately $(\pi/4)\sqrt{(N/M)}$ iterations for M solutions among N candidates [15, 16].

2.4 Gates, Universality and Compilation

Computation proceeds by unitary evolution. A universal set is obtained from arbitrary single-qubit rotations together with one entangling two-qubit gate, conventionally CNOT or CZ. Practical compilation, however, is governed by discrete gate sets, and the Solovay–Kitaev theorem guarantees that any single-qubit unitary can be approximated to accuracy ϵ using $O(\log^c(1/\epsilon))$ gates from a suitable discrete set, with c slightly greater than one [14]. This polylogarithmic overhead is benign in isolation but compounds badly when combined with the routing cost of limited connectivity.

Contemporary two-qubit fidelities cluster around 99.921% for trapped ions [8] and roughly 99.5–99.9% for superconducting devices [11]. Circuit depth remains the binding constraint on unencoded hardware: because errors accumulate multiplicatively, a circuit of depth d with per-gate error ϵ retains useful signal only while $d\epsilon$ remains small, which bounds present devices to a few hundred layers before the output is indistinguishable from noise.

2.5 Complexity-Theoretic Framing

It is worth stating plainly what is and is not proven. The class BQP contains problems solvable in polynomial time by a quantum computer with bounded error; it is known to contain integer factorisation and discrete logarithm, and is believed — but not proven — to lie outside P. No proof exists that $BQP \neq BPP$, and consequently every claim of exponential quantum advantage rests on the presumed classical hardness of the underlying problem rather than on an unconditional separation. This is not a pedantic caveat: it is the reason the Bennett–Bernstein–Brassard–Vazirani result, which shows that unstructured search admits no better than quadratic quantum improvement [17], carries greater evidential weight than most positive claims. Structure, not superposition, is the source of speedup.

The prevailing operating regime remains the noisy intermediate-scale quantum (NISQ) regime characterised by Preskill: hundreds to thousands of physical qubits, no error correction, and circuit depths limited by decoherence [3]. The field's central engineering question is how quickly this regime can be exited.

3. Hardware Modalities

3.1 Evaluation Criteria

Qubit count is the most frequently quoted and least informative single metric. A defensible comparison requires at minimum the criteria set out by DiVincenzo — a scalable register of well-characterised qubits, reliable initialisation, coherence long relative to gate duration, a universal gate set, and high-fidelity readout [2] — supplemented by four engineering considerations that determine achievable circuit volume:

- **Connectivity.** All-to-all coupling eliminates SWAP insertion. On a fixed-lattice device, routing can inflate two-qubit gate count by an order of magnitude for dense circuits, so nominal fidelity overstates effective fidelity.

- **Gate speed.** Ion traps operate on microsecond timescales; superconducting gates run in tens of nanoseconds. A three-order-of-magnitude clock-rate difference dominates wall-clock throughput even where fidelity favours the slower platform.
- **Composite benchmarks.** Quantum Volume compresses width, depth and fidelity into a single figure; circuit-layer operations per second (CLOPS) captures throughput; error per layered gate (EPLG) measures degradation under realistic parallel load rather than in isolation [34].
- **Infrastructure burden.** Dilution refrigeration, ultra-high vacuum, and laser systems impose capital and operating costs that scale differently across modalities and that are rarely reported alongside performance figures.

The comparison that follows applies these criteria consistently. It should be read with the caveat that most figures are vendor-reported under vendor-chosen conditions.

3.2 Superconducting Circuits

Transmon qubits are lithographically fabricated Josephson-junction circuits operated at 15–20 mK in dilution refrigerators, with single-qubit control effected by shaped microwave pulses in the 4–8 GHz band [18]. The modality benefits from semiconductor process maturity and the fastest gate times of any platform.

The 156-qubit Heron R2 generation adopts a heavy-hexagonal lattice with tunable couplers, a topology chosen deliberately to trade connectivity for crosstalk suppression [5, 19], and reports error per layered gate of about 2.15×10^{-3} at the hundred-qubit scale [11]. The competing 105-qubit Willow device demonstrated below-threshold surface-code operation in December 2024, with logical error rate suppressed by roughly a factor of two at each increment of code distance [20, 21] — the first experimental confirmation that adding physical qubits to an encoded memory improves rather than degrades it. Median relaxation times of order 500 μ s are now reported at the state of the art [22].

The constraints are structural rather than incidental. Fixed planar connectivity imposes routing overhead; the cryogenic plant and coaxial control wiring scale with qubit number and represent the principal obstacle to million-qubit systems; and fabrication variability across a wafer produces qubit-to-qubit parameter spread that must be handled in calibration.

3.3 Trapped Ions

Ions confined in radio-frequency Paul traps store logical information in internal electronic states, with gates driven by laser or microwave fields mediated through shared motional modes. The Helios system provides 98 fully connected qubits at a reported two-qubit gate fidelity of 99.921% — the highest published figure across all modalities [8]. Coherence exceeds ten minutes for single qubits and, in decoherence-free subspaces, has been extended to the hour scale [23, 24].

All-to-all connectivity is a substantial and often underweighted advantage: it removes SWAP overhead entirely, so a trapped-ion device of a given nominal size executes dense circuits that would require considerably more resources on a fixed-lattice machine. Against this, gate durations of roughly 1–100 μ s place the platform three orders of magnitude behind superconducting circuits in raw clock rate, and confining more than about a hundred ions in a single trap becomes unstable. Scaling strategies therefore centre on quantum charge-coupled device architectures in which ions are shuttled between interaction and storage zones, or on photonic interconnects between separate traps — both add engineering complexity that has not yet been demonstrated at scale.

3.4 Photonic Systems

Photonic processors encode information in optical modes and manipulate it through beam splitters, phase shifters and interferometric networks. The Borealis machine generates 216 squeezed-state modes entangled in three dimensions [25], and, because photons interact negligibly with their environment during propagation, operates at room temperature with essentially no decoherence in transit.

Two problems are fundamental rather than engineering. Linear-optical two-qubit gates are inherently probabilistic and succeed only with heralded measurement outcomes, so deterministic operation demands multiplexing and feed-forward. Photon loss, meanwhile, is not correctable by conventional means and accumulates through every component. Fusion-based architectures targeting a million qubits address both by measuring small entangled resource states and stitching successful fusions into a fault-tolerant cluster [26]; the approach is theoretically attractive but the component-level loss budgets it requires remain demanding. Detector efficiency and single-photon source purity, rather than qubit count, are the binding constraints.

3.5 Neutral Atoms

Optical tweezer arrays hold individual atoms in reconfigurable geometries and entangle them by excitation to Rydberg states, where dipole–dipole interaction produces a blockade effect [27]. The modality has scaled fastest in the recent period: systems now offer more than 1,200 qubits with effectively arbitrary connectivity, achieved by physically transporting atoms between gate zones, and a coherence time of 40 seconds obtained by encoding in the nuclear spin of strontium [9, 28]. Parallel two-qubit gates at approximately 99.5% fidelity have been demonstrated across 60 atoms simultaneously [29].

The combination of large registers, reconfigurable connectivity and room-temperature vacuum operation makes this the most rapidly improving platform, and it now hosts several of the most advanced logical-qubit demonstrations. Its weaknesses are atom loss during transport, gate fidelities still short of trapped ions, and mid-circuit measurement that is harder to perform without disturbing neighbouring sites.

3.6 Quantum Annealing

Annealers evolve a system adiabatically from the ground state of a simple Hamiltonian to that of a problem Hamiltonian encoding a quadratic unconstrained binary optimisation instance. The Advantage2 architecture provides more than 4,400 qubits with twenty-way connectivity in a Zephyr topology [30], the largest register of any commercial system.

The device is not a universal quantum computer and cannot execute arbitrary circuits, so its qubit count is not commensurable with gate-model figures. Whether annealing delivers genuine speedup over well-tuned classical heuristics remains contested: results are sensitive to problem class, embedding overhead and the choice of classical baseline, and comparisons against simulated annealing or parallel tempering have frequently narrowed or eliminated the reported advantage. The platform’s realistic value lies in rapid heuristic sampling for structured combinatorial problems rather than in asymptotic advantage.

3.7 Topological Qubits

The topological approach encodes information non-locally in Majorana zero modes localised at the ends of a semiconductor nanowire proximitised by a superconductor, on the argument that non-local encoding confers intrinsic protection against local noise [31]. Gate-defined tetron devices in the Majorana 2 generation report a thousandfold reliability improvement over the preceding device, mean state lifetimes of about 20 seconds and microsecond operations [31], with eight qubits integrated on a die architected for eventual million-qubit scaling [32].

This remains the least mature modality and the most scientifically contested. Interpretation of the tunnelling spectroscopy signatures used to identify Majorana modes has been challenged in the literature, and distinguishing genuine topological states from trivial Andreev bound states producing similar signatures is an open experimental problem [33]. The engineering upside is real — hardware-level protection would substantially reduce error-correction overhead — but the evidentiary bar has not yet been cleared.

3.8 Comparative Synthesis

Figure 1 plots reported register size on a logarithmic axis with the corresponding two-qubit fidelity annotated alongside. The inverse relationship between the two is the central fact of the current hardware landscape.

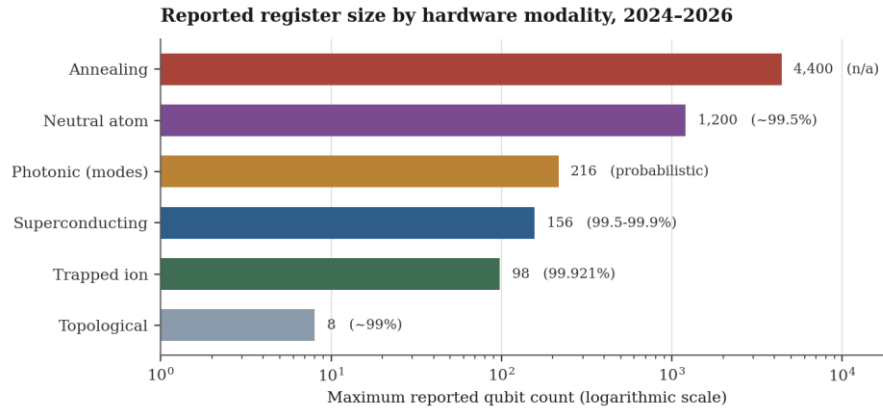


Figure 1. Maximum reported qubit count by modality, annotated with two-qubit gate fidelity. Annealing qubits are not commensurable with gate-model qubits and photonic figures count optical modes rather than logical qubits; the axis compares magnitude, not capability.

Table 1 consolidates the reported characteristics of the six modalities. Read across rather than down: no platform leads on every axis, and the leading figure in each column belongs to a different technology.

Modality	Qubits	Coherence	2Q Fidelity	Connectivity	Principal Constraint	Representative Developers
Superconducting	105–156	$T_1 \approx 500 \mu\text{s}$	99.5–99.9%	Fixed lattice	Cryogenic and wiring scale-up	IBM, Google, Rigetti
Trapped ion	56–98	Minutes to hours	99.921%	All-to-all	Slow gates; trap size limit	Quantinuum, IonQ
Photonic	216 modes	Loss-limited	Probabilistic	Reconfigurable	Photon loss; heralding	Xanadu, PsiQuantum
Neutral atom	1,200+	$\approx 40 \text{ s}$	$\approx 99.5\%$	Reconfigurable	Atom loss; mid-circuit readout	Atom Computing, QuEra, Pasqal
Annealing	4,400+	Not applicable	Not applicable	20-way (Zephyr)	Non-universal; contested speedup	D-Wave
Topological	≈ 8	$\approx 20 \text{ s}$	$\approx 99\%$	Under development	Unresolved materials physics	Microsoft

Table 1. Comparative characteristics of quantum hardware modalities, 2024–2026. Figures are as reported by developers under conditions of their own choosing and are not independently benchmarked; cross-platform comparison should be treated as indicative.

Two observations follow. First, the ordering by qubit count is close to the inverse of the ordering by fidelity, which is the expected signature of a field in which scale and control quality are traded against one another. Second, connectivity differences are large enough to invert naive comparisons: a fully connected 98-qubit register can execute dense circuits that a 156-qubit fixed-lattice device cannot, once routing overhead is accounted for. Any claim of platform superiority that rests on a single column of Table 1 should be treated as marketing rather than measurement.

4. Algorithmic Landscape

Quantum algorithms may be grouped by the strength of the guarantee they carry. A first group possesses proven asymptotic separation against the best known classical method, conditional on standard hardness assumptions. A second group possesses proven optimality within a restricted model. A third — by far the largest in current practice — is heuristic, offering no performance guarantee and requiring empirical validation problem by problem. Conflating these categories is the most common source of overstatement in the applied literature.

4.1 Shor’s Algorithm and Integer Factorisation

Shor’s algorithm converts factoring into period-finding. For a composite N one selects a random $a < N$, prepares a superposition over exponents, evaluates $f(x) = a^x \bmod N$ in superposition, and applies the quantum Fourier transform to extract the period r of f by phase estimation. Non-trivial factors then follow classically from $\gcd(a^{r/2} \pm 1, N)$ [35, 36]. The gate complexity is $O(n^3)$ for an n -bit modulus [37], against the sub-exponential running time of the general number field sieve [38] — an exponential separation in the asymptotic sense, though the practical crossover point is governed entirely by constant factors and error-correction overhead.

Resource estimation has moved sharply in one direction. A 2019 analysis placed RSA-2048 at roughly twenty million physical qubits and about eight hours of runtime [39]. Optimised approaches published since suggest that fewer than one million noisy physical qubits may suffice, factoring the same modulus in under a week [40, 98] — a reduction of approximately twenty-fold achieved through improved modular arithmetic, magic-state cultivation and more efficient use of idle qubits, with no change in the underlying algorithm. Because the improvement came from compilation rather than hardware, further reductions of the same kind cannot be excluded, and this asymmetry is the strongest argument for treating migration deadlines as conservative.

Experimental progress remains far behind. The largest integer factored on genuine quantum hardware without classical pre-processing that trivialises the instance is 48 bits, using ten superconducting qubits [41]; a generalised period-decomposition method has reported success rates above 99.998% for seven-digit numbers on noisy devices [42]. The gap between 48 bits and 2048 bits is not incremental, and claims of larger factorisations should be scrutinised for whether the quantum processor performed the computationally hard step or merely a decorative fragment of it.

4.2 Grover’s Algorithm and Amplitude Amplification

Grover’s search provides a quadratic speedup for unstructured search, requiring approximately $(\pi/4)\sqrt{N/M}$ oracle calls [15, 16]. Unlike Shor’s exponential separation, this bound is provably optimal: the BBBV theorem establishes that no quantum algorithm can beat $O(\sqrt{N})$ for a genuinely unstructured oracle [17].

The cryptographic implication is frequently overstated. For AES-256 key recovery, amplitude amplification reduces roughly 2^{256} classical queries to roughly 2^{128} quantum queries [39]. Two considerations blunt this. First, the quadratic speedup is destroyed in practice by error-correction overhead, since each oracle call must be executed fault-tolerantly and the resulting circuits are extraordinarily deep. Second, Grover parallelises poorly — running k machines yields only a $\sqrt{k}$ improvement — so the attack cannot be shortened by throwing hardware at it. The consensus that symmetric cryptography requires only a doubling of key length, rather than replacement, follows from these facts [39].

4.3 Variational Quantum Eigensolvers

The variational quantum eigensolver prepares a parameterised trial state on quantum hardware, measures the expectation value of a problem Hamiltonian, and delegates parameter updates to a classical optimiser in an iterative loop [43]. Its appeal for near-term hardware is that circuit depth stays shallow and systematic errors are partially absorbed into the variational parameters. The variational principle guarantees that the measured energy is an upper bound on the true ground-state energy, which provides a one-sided correctness certificate [44].

Demonstrations have reached molecules of the scale of ethylene [45]. Scaling is obstructed by barren plateaus — regions of parameter space in which the gradient variance decays exponentially with qubit number, rendering optimisation infeasible because the number of measurement shots required to resolve a gradient grows exponentially [46, 47]. Barren plateaus are now understood to arise from several distinct mechanisms, including excessive ansatz expressibility, entanglement between the register and traced-out subsystems, and hardware noise itself, and each mechanism resists a different remedy. A further difficulty is that the measurement cost of estimating a molecular Hamiltonian’s expectation value grows steeply with orbital count, so the classical optimisation loop, not the quantum circuit, often dominates total runtime.

4.4 Quantum Approximate Optimisation

The quantum approximate optimisation algorithm encodes a combinatorial objective into alternating applications of a problem unitary and a mixing unitary over p layers, giving $2p$ variational parameters tuned classically [48]. Under the fixed-angle conjecture, QAOA is projected to exceed the Goemans–Williamson approximation ratio on 3-regular graphs at $p \geq 11$ [49]; utility-scale application is considered plausible when combined with error mitigation [50], and experiments have been run on up to 55 qubits [51].

Two cautions apply. The depth required for the conjectured advantage exceeds what current hardware sustains at useful width, so the projection is not yet testable at scale. More seriously, the relevant comparison is not against a naive classical baseline but against the best available classical solver, and on that comparison the evidence has been unfavourable — a benchmark study spanning 250 portfolio-optimisation instances found every instance solved to proven optimality by classical mixed-integer programming within seconds [10]. QAOA’s prospects rest on identifying problem classes where classical methods degrade sharply, and such classes have proven surprisingly difficult to find.

4.5 Hamiltonian Simulation

Simulation of quantum systems is the application for which quantum computers were originally proposed and remains the most defensible near-term target, because the mapping from physical system to quantum register is natural rather than contrived. Principal techniques include Trotter–Suzuki product formulas, linear combination of unitaries, and qubitisation, all used to estimate ground-state energies governing molecular stability and reactivity [52, 53]. Rigorous analysis has shown Trotter error to be substantially

smaller in practice than worst-case bounds suggest, particularly for local Hamiltonians with commuting structure, which materially lowers projected resource requirements [56].

Compilation advances have compounded this. Modifications to the Jordan–Wigner transformation reduce implementation cost to a constant in the number of orbitals [54], and combining Wannier-basis Hamiltonians with hybrid fermion-to-qubit mappings yields circuits whose depth is independent of system size [55].

Two 2026 results anchor the empirical case. In April 2026, an echo-based protocol on the Willow processor produced the first verifiable demonstration of quantum advantage, verifiability being the salient property — earlier sampling-based claims were repeatedly eroded by improved classical simulation because their outputs could not be independently checked [4]. In July 2026, a collaboration demonstrated quantum advantage in simulating heterogeneous quantum materials within a framework designed for trusted computation beyond classical verification [12]. Both concern scientific rather than commercial advantage, and neither yet corresponds to a problem an enterprise would pay to solve.

4.6 Classical Competition and Dequantisation

Any assessment of quantum algorithms must account for the moving classical baseline. A line of work initiated by quantum-inspired classical sampling algorithms for recommendation systems demonstrated that several claimed exponential speedups in quantum machine learning survived only because the classical alternative had not been examined under matching data-access assumptions; when granted analogous sample-and-query access, classical algorithms recovered polynomial-time performance [57]. Comparable erosion has occurred in random-circuit sampling, where successive tensor-network methods have narrowed or closed previously announced gaps.

The practical lesson is procedural rather than pessimistic: a quantum advantage claim is only as strong as the classical baseline it is measured against, and that baseline is not static. This is precisely why verifiable advantage protocols represent a methodological advance over sampling-based claims.

Algorithm	Nature of Speedup	Guarantee	Experimental Status	Primary Domain
Shor	Exponential	Proven, conditional on classical hardness	48-bit factored; RSA scale remote	Cryptanalysis
Grover	Quadratic	Proven optimal (BBBV)	Small demonstrations only	Search, symmetric cryptanalysis
VQE	None proven	Heuristic; energy upper bound only	Molecules to ethylene scale	Quantum chemistry
QAOA	None proven	Heuristic; conjectural at $p \geq 11$	Up to 55 qubits	Combinatorial optimisation
Hamiltonian simulation	Exponential for local Hamiltonians	Proven for defined classes	Verifiable advantage, 2026	Materials and chemistry
Quantum kernel methods	Case-dependent	Contested; dequantisation risk	Small-scale pilots	Machine learning

Table 2. Algorithmic families classified by strength of performance guarantee. The distinction between proven and heuristic speedup is the single most important axis for assessing commercial claims.

5. Software Frameworks and Access Models

5.1 Programming Frameworks

The software layer has consolidated around five substantial frameworks, each reflecting a different view of where the abstraction boundary should sit.

Qiskit 1.0, released in February 2024, marked a shift from rapid feature growth to interface stability, delivering a 55% reduction in memory consumption relative to version 0.39 and approximately sixteenfold faster binding and transpilation relative to version 0.33 [58]. The SDK supports circuits beyond a hundred qubits and is architected for workloads an order of magnitude larger [59]. Execution is mediated by Runtime primitives — Sampler and Estimator — which abstract hardware-specific complexity and support Eagle, Heron and Falcon generations [60]. The primitive abstraction is significant beyond convenience: it lets error mitigation be applied transparently at the service layer rather than being reimplemented per application.

Cirq is designed around explicit hardware control for NISQ-scale devices of order one hundred qubits executing of order one thousand gates [61], and serves as the principal interface to Sycamore and Willow processors [62]. Built-in simulators support state-vector and density-matrix evolution with noisy channels sampled by Monte Carlo methods [63], and the moment structure of a circuit maps directly onto hardware scheduling, giving precise control of gate timing [64]. It is the appropriate choice when pulse-level timing matters and the least convenient when it does not.

Q# and the Azure Quantum Development Kit are differentiated by the Resource Estimator, which projects qubit counts and runtime for a specified target architecture before any hardware is consumed [65]. Open-sourced in early 2024 with performance improvements up to one hundredfold [66], it exposes six predefined qubit parameter models and supports both surface-code and Floquet-code assumptions [67]. Given that most fault-tolerant algorithms cannot be run today, resource estimation is arguably the most immediately useful capability in the entire toolchain.

PennyLane extends automatic differentiation across quantum and hybrid computation [68] and integrates with PyTorch, TensorFlow and JAX. Its plugin architecture permits an unmodified circuit to execute on multiple vendors' backends [69], and it is generally regarded as the first framework purpose-built for quantum machine learning [70].

QuTiP 5.0 simulates open- and closed-system dynamics via master equations and Monte Carlo wavefunction methods [71]. It provides no hardware execution path and is therefore a theoretical and pedagogical instrument rather than a deployment framework [72] — a limitation, but also the reason it remains the reference implementation for open-system dynamics.

A structural weakness of the ecosystem deserves note. Circuit-level portability across frameworks is now routine, but performance portability is not: a circuit transpiled for a heavy-hexagonal lattice performs poorly on an all-to-all machine and conversely, so cross-platform benchmarking remains labour-intensive and comparisons published by vendors are rarely reproducible without substantial re-optimisation.

5.2 Cloud Access and Commercial Models

Four access models dominate, and they differ in kind rather than degree. The first offers the largest fleet of superconducting processors under tiered pricing: a free plan capped at ten minutes per month, pay-as-you-go at \$1.60 per second, and premium access at approximately \$48 per minute [73, 74, 75]. The second operates as a multi-vendor marketplace with no upfront commitment and per-task plus per-shot charges, integrated with a broader cloud platform [76, 77, 78]. The third provides \$500 in credits per hardware provider for first-time users and brokers access to trapped-ion systems [79, 80, 81]. The fourth is not

commercial at all but a research-access programme under which investigators submit proposals specifying circuits and observables [82, 83].

The pricing structure carries a strategic implication that is easy to miss. At \$1.60 per second, a variational optimisation requiring tens of thousands of circuit executions becomes expensive quickly, and the cost is dominated by the classical optimisation loop’s repeated hardware calls rather than by any single quantum computation. Shot-efficient algorithm design is therefore an economic requirement, not merely an aesthetic one.

Access Model	Hardware Exposed	Commercial Terms	Differentiator
Largest superconducting fleet	Eagle, Heron, Falcon, Nighthawk	\$1.60/s pay-as-you-go; $\approx$ \$48/min premium	Runtime primitives; integrated error mitigation
Multi-vendor marketplace	IonQ, Rigetti, QuEra, IQM	Per-task plus per-shot	Cross-modality comparison in one environment
Brokered trapped-ion access	IonQ, Quantinuum	\$500 credits per provider	Resource Estimator for pre-hardware planning
Research-access programme	Willow	Proposal-based, non-commercial	Error-correction leadership

Table 3. Quantum cloud access models. The divergence between commercial and research-access strategies reflects differing judgements about whether the field is closer to a product or to a laboratory instrument.

6. Cryptographic Implications and Post-Quantum Migration

6.1 Quantum Key Distribution

Quantum key distribution derives security from physical law rather than computational hardness, and is therefore secure against adversaries of unbounded computational power, quantum or classical [84]. The BB84 protocol encodes random bits in conjugate photon polarisation bases; the no-cloning theorem ensures that interception necessarily perturbs the transmitted states and is detectable in the error rate [85]. The E91 protocol achieves the same end through violation of a Bell inequality on entangled pairs, transferring the security argument from state disturbance to correlation statistics [86]. Practical systems employ decoy-state BB84 to close the photon-number-splitting vulnerability introduced by attenuated laser pulses [87].

Deployment has reached operational scale. A national network exceeding 10,000 kilometres integrates 145 fibre backbone nodes and 20 metropolitan networks across 17 provinces and 80 cities, linked by satellite for space-to-ground distribution [88, 89]. A European initiative connects all 27 member states through combined terrestrial and satellite segments [90], and BB84 operation has been demonstrated over 303 kilometres of deployed fibre [85]. Commercial deployments include a network securing distributed-ledger infrastructure [91].

The limitations are structural and rarely stated with sufficient force. Quantum key distribution solves key agreement only; it does not provide authentication, which must still be established by classical means, and it therefore does not by itself defeat an active man-in-the-middle adversary. It requires dedicated optical infrastructure, degrades with distance in the absence of trusted relays or as-yet-undeployed quantum repeaters, and does nothing for data at rest or for code signing. Several national technical authorities have consequently declined to recommend it as a substitute for post-quantum cryptography. The two technologies address different threat surfaces and are complements at best.

6.2 Standardised Post-Quantum Primitives

The first three post-quantum standards were published on 13 August 2024 [6]: ML-KEM (FIPS 203) for key encapsulation, based on the module learning-with-errors problem [92]; ML-DSA (FIPS 204) for digital signatures over module lattices [93]; and SLH-DSA (FIPS 205), a stateless hash-based signature scheme whose security rests only on the properties of the underlying hash function and which therefore offers the most conservative assumptions available [94]. Further primitives are in progress: FN-DSA (FIPS 206), an NTRU-lattice scheme with signatures smaller than ML-DSA, submitted for approval in August 2025 [95], and HQC, a code-based key encapsulation mechanism selected in March 2025 as a structurally independent backup to ML-KEM [96].

The rationale for maintaining backups across distinct mathematical families is not redundancy for its own sake. Lattice-based schemes share a common hardness assumption; a cryptanalytic advance against module lattices would compromise ML-KEM and ML-DSA simultaneously. A code-based KEM and a hash-based signature scheme fail independently, which is the property that matters.

Performance characteristics favour adoption rather than obstructing it. ML-KEM-512 completes a full key exchange in 35.7 ms on an ARM Cortex-M0+, approximately seventeen times faster than ECDH over P-256 on equivalent hardware [97]. The migration cost lies not in computation but in the larger key and ciphertext sizes, which stress constrained protocols, embedded devices and certificate chains — the practical bottleneck is bandwidth and protocol rigidity, not CPU time.

6.3 Threat Timeline and Migration Mandates

Estimates of when a cryptographically relevant quantum computer will exist have compressed sharply. Analysis published in May 2025 demonstrated that fewer than one million noisy physical qubits could factor RSA-2048 in under a week, roughly a twentyfold reduction against the twenty-million-qubit estimate of 2019 [98]. Independent assessments assign a 14–34% probability to a cryptographically relevant machine emerging by 2033–2035 [99], with one analysis placing the most likely date for offline attacks against ECC-256 at 2032 [100].

The operative risk, however, is not the arrival date. Harvest-now-decrypt-later attacks require only that an adversary record ciphertext today and retain it until decryption becomes feasible; 87% of surveyed organisations report concern about such scenarios [101]. For any data with a confidentiality lifetime exceeding the interval to a capable machine, exposure has already occurred. Mosca’s inequality formalises the point: if the time data must remain secret plus the time required to migrate exceeds the time until a capable adversary exists, the organisation is already late.

Mandates have accordingly hardened from guidance into schedule. Federal guidance deprecates quantum-vulnerable algorithms from 2030 and disallows them from 2035 [102, 103]; CNSA 2.0 requires new national-security acquisitions to support post-quantum cryptography from January 2027 [104]; and an executive order sets binding 2030–2031 deadlines for federal high-value assets [105]. At least one major technology provider has committed to completing internal migration by 2029, with national targets of 2035 and a G7 target of 2034 [101].

Complementary frameworks address the mechanics. ETSI TR 104 016 defines a repeatable methodology for quantum-safe migration [106], and TS 103 744 specifies hybrid key establishment combining classical and post-quantum mechanisms during transition [107]. Hybrid construction is the pragmatic consensus: composing an established elliptic-curve exchange with ML-KEM means an implementation flaw or cryptanalytic break in the newer primitive does not by itself compromise the session. Hybrid key agreement

has already been deployed at internet scale in major browsers and content-delivery networks, making it the most widely exercised post-quantum construction in production [108].

The practical implication for organisations is that cryptographic inventory — knowing where and how public-key cryptography is used across an estate, including in embedded firmware, code-signing chains and third-party dependencies — is the gating task, and it is typically slower than the cryptographic substitution itself. Crypto-agility, the capacity to replace primitives without re-architecting, is the durable objective; the current migration is best understood as the first exercise of that capability rather than a one-off event.

7. Industry Applications and Quantum Machine Learning

Most industrial quantum activity sits between Technology Readiness Levels 3 and 6 — experimental proof of concept through demonstration in a relevant environment — and commercial advantage on production-scale problems remains several years away. The sections below distinguish between results that constitute genuine scientific output and results that constitute feasibility exploration.

7.1 Chemistry, Drug Discovery and Life Sciences

Molecular simulation is the most defensible near-term application because the target system is itself quantum mechanical, so the mapping to a quantum register is natural rather than engineered. A quantum-centric supercomputing framework combining quantum processors with classical high-performance computing was used to compute the electronic structure of protein–ligand complexes at 12,635 atoms, the largest biologically meaningful molecular system addressed with quantum hardware to date [109]. The architecture matters as much as the result: quantum resources handled the strongly correlated active space while classical resources handled the remainder, which is the realistic deployment pattern for the coming decade.

Resource-estimation work using a fault-tolerant photonic architecture projected 234-fold and 278-fold computational speedups for Cytochrome P450 and the FeMoco cluster respectively [110]. These are projections contingent on machines that do not yet exist, and should be read as scoping studies rather than measurements. Active industrial programmes include exploration of mRNA medicine design [111] and molecular simulation partnerships targeting problems beyond classical tractability [112].

A candid assessment: for the great majority of pharmaceutically relevant molecules, density functional theory and coupled-cluster methods remain adequate and vastly cheaper. The quantum case is confined to strongly correlated systems — transition-metal catalytic centres, certain excited-state problems, nitrogenase-like clusters — where classical methods degrade. That is a genuine and valuable niche, but it is a niche.

7.2 Financial Services

Financial applications have produced both the sector’s most substantial programmes and its clearest negative result. A decomposition pipeline developed jointly by a major bank and a cloud provider reduced real-world portfolio-optimisation problems by approximately 80% while preserving solution quality, achieving a threefold reduction in time to solution for instances of up to 1,500 variables [113]. A hybrid algorithm solved an optimisation over 225 financial assets using circuits of up to 78 qubits and more than 1,000 two-qubit gates on trapped-ion hardware [114].

Set against this, resource estimation for a key portfolio problem at another major institution indicated a requirement of at least eight million logical qubits and computation times measured in millions of years —

not a hardware-generation gap but a categorical one — leading the firm to dismantle its quantum team in 2026 [13]. The systematic benchmark cited earlier reinforces the point: across 250 portfolio instances, classical mixed-integer programming reached proven optimality in seconds [10].

The reconciliation is instructive. The successful results are hybrid and use quantum resources for a narrow sub-step within a classically orchestrated pipeline; the negative results arise where a full problem was mapped monolithically onto quantum hardware. Where quantum methods have added value in finance, the value has come from problem decomposition — an insight that is itself partly classical and portable.

7.3 Materials Science and Logistics

Quantum simulation of real magnetic materials produced results matching neutron scattering measurements in March 2026 [115], a validation against independent physical experiment rather than against another simulation — methodologically the strongest form of evidence available in this area. Battery research applies quantum embedding to electrode surface reactions [116], and automotive catalyst modelling targets strongly correlated transition-metal oxides [117], both of which fall squarely within the strongly correlated niche identified above.

In logistics, a hybrid solver achieved a 3.86% optimality gap on capacitated vehicle routing with time windows in a study supported by a commercial freight operator, though time-window feasibility degraded as instance size grew [118]. A 3.86% gap is respectable but not competitive with mature classical routing heuristics on comparable instances, and the degradation with scale is the more significant finding.

7.4 Quantum Machine Learning

Quantum machine learning employs variational quantum circuits as parameterised models, and inherits the barren plateau problem in an acute form: gradient variance vanishes exponentially in qubit count or circuit depth [47]. The literature has identified a structural tension — deep circuits suffer vanishing gradients while shallow circuits encounter poor local minima — which constrains architecture search from both directions [119]. Mitigation strategies under investigation include modified data encodings, layerwise training, correlated parameter initialisation and trainability-preserving architectures [120]. Applications explored to date span fraud detection, portfolio optimisation and market prediction via quantum kernel methods [121], with dedicated classical models being developed to improve quantum circuit performance [122].

Theoretical work has clarified when a quantum advantage in learning is even possible. Analyses of the interaction between classical data access and quantum models indicate that for many tasks a classical learner with sufficient training data matches quantum performance, and that advantage requires the data-generating process itself to be quantum in origin [123]. Combined with dequantisation results [57], this suggests that quantum machine learning on classical datasets is the least promising branch of the field, while learning on data produced by quantum sensors, experiments or simulations is the most promising. Present evidence does not support claims of near-term advantage on conventional business datasets.

8. Quantum Error Correction and Systemic Constraints

8.1 Error Mechanisms

Quantum states are fragile in ways that classical bits are not. Amplitude damping (characterised by T_1) reflects energy loss to the environment; dephasing (T_2) reflects randomisation of relative phase through coupling to thermal noise, electromagnetic interference and charge fluctuation. Even the best current

devices experience at least one failure per thousand operations, with two-qubit gate error typically five to ten times single-qubit gate error [124].

Two error classes are more troublesome than raw rates suggest. Crosstalk — residual coupling between nominally idle qubits — can push surface-code architectures from a computationally favourable regime into an error-dominated one, because correlated errors violate the independence assumptions on which threshold calculations rest [125]. Readout errors arising from imperfect state discrimination propagate through syndrome extraction cycles and can be misinterpreted by the decoder as data errors [126]. Both are correlated rather than stochastic, and correlated noise is precisely what error-correcting codes handle worst.

8.2 The Surface Code and Its Overhead

The surface code arranges physical qubits on a $d \times d$ lattice in which data qubits hold the encoded state and ancilla qubits extract error syndromes through repeated stabiliser measurement. Code distance d determines correction capacity at $\lfloor (d-1)/2 \rfloor$ simultaneous errors [124]. Its practical dominance derives from requiring only nearest-neighbour connectivity and from a comparatively high threshold of roughly 0.7–1% under circuit-level noise [127], which places it within reach of existing hardware — the threshold theorem guarantees arbitrarily long computation once physical error rates fall below this value, at polylogarithmic overhead [140].

The cost is severe. At physical error rates near 10^{-3} , roughly 1,000 to 2,000 physical qubits are required per logical qubit for classically intractable algorithms [128]. This ratio, not qubit count, is the true measure of distance to fault tolerance: a million-physical-qubit machine at this overhead yields only hundreds to a thousand logical qubits. It also explains why the resource-estimation reductions discussed in Section 4.1 were so consequential — they attacked the overhead rather than the register size.

8.3 Landmark Demonstrations

Progress from 2021 to 2025 followed a coherent trajectory: repetition codes on 21 qubits in 2021 [129]; a distance-3 surface code on 17 qubits in 2022 [130]; 48 logical qubits realised across 280 neutral atoms in 2023 [131]; quantum low-density parity-check codes achieving roughly tenfold efficiency over surface codes on 288 qubits in 2024 [127]; an 800-fold error improvement on 30 trapped-ion qubits in 2024 [132]; below-threshold operation on 105 superconducting qubits in 2024 [133]; and a universal fault-tolerant architecture across 448 neutral atoms in 2025 [134].

The December 2024 below-threshold result warrants specific attention. A distance-7 code on 101 physical qubits achieved a logical error rate of $0.143\% \pm 0.003\%$ per cycle, with each increase in lattice size from 3×3 to 5×5 to 7×7 suppressing errors by a factor $\Lambda = 2.14 \pm 0.02$ [133, 124]. The significance is qualitative rather than quantitative: it established that the encoded memory improves as it grows, which is the defining property of a scalable fault-tolerant architecture and had never previously been observed. Complementary results include four logical qubits from only 30 physical trapped-ion qubits with error rates 800 times below the corresponding physical rates [135], and the first demonstration of magic-state distillation at the logical level on neutral atoms [136] — the latter being necessary because transversal gate sets are provably incomplete and non-Clifford operations must be supplied by distillation or an equivalent mechanism.

Year	Demonstration	Scale	Significance
2021	Repetition code	21 qubits	Exponential error suppression in one dimension
2022	Distance-3 surface code	17 qubits	First full surface-code cycle

Year	Demonstration	Scale	Significance
2023	Logical qubit array on neutral atoms	48 logical / 280 atoms	Logical operations at scale
2024	qLDPC (bivariate bicycle) codes	12 logical / 288 qubits	$\approx 10\times$ overhead reduction vs. surface code
2024	Trapped-ion logical encoding	4 logical / 30 qubits	$800\times$ improvement over physical error rate
2024	Below-threshold surface code	$d = 7$, 101 qubits	$\Lambda = 2.14$; error falls as code grows
2025	Universal fault-tolerant architecture	448 atoms	Universal logical gate set demonstrated
2025	Logical magic-state distillation	Neutral atoms	Non-Clifford resource at logical level

Table 4. Milestones in quantum error correction, 2021–2025. The sequence shows a transition from demonstrating that codes function to demonstrating that they improve with scale.

8.4 Alternative Code Families

The surface code’s overhead has motivated structurally different constructions. Bivariate bicycle quantum low-density parity-check codes preserve 12 logical qubits for nearly one million syndrome cycles using 288 physical qubits, against approximately 3,000 required by surface codes for comparable performance — close to a tenfold efficiency gain [127, 137]. The trade-off is that these codes require longer-range connectivity than a planar nearest-neighbour lattice provides, which suits some hardware modalities considerably better than others.

Colour codes offer native transversal gates and potentially fewer physical qubits per logical qubit, at the cost of a lower threshold [138]. The Tesseract code combined fault-tolerant error correction with logical operations on 12 logical qubits for the first time [139]. A further direction encodes information in bosonic modes, where the dominant error channel is biased toward a single type and can be suppressed at the hardware level, reducing the burden placed on the outer code; results in this family have shown substantial reductions in the physical resources required per logical qubit [141]. Taken together, these developments indicate that the surface code should be regarded as a well-characterised baseline rather than the destination.

8.5 The Path to Fault Tolerance

Useful algorithms require logical error rates in the range of 10^{-10} to 10^{-15} [128] — seven to twelve orders of magnitude below current physical rates, a gap that must be closed by encoding rather than by improved components alone. Published targets include a fault-tolerant system running 100 million quantum gates on 200 logical qubits using qLDPC codes by 2029 [142], and a universal fault-tolerant machine by 2030 [143].

In the interim, error mitigation serves as a bridge. Zero-noise extrapolation runs a circuit at deliberately amplified noise levels and extrapolates to the zero-noise limit; probabilistic error cancellation inverts a characterised noise channel by sampling; dynamical decoupling inserts pulse sequences that refocus low-frequency dephasing during idle periods [126, 144, 145, 146]. These techniques extract usable results from noisy hardware but do not scale: the sampling overhead of probabilistic error cancellation grows exponentially with circuit volume. Mitigation buys time; it does not substitute for correction [147].

8.6 Systemic Obstacles Beyond Error Correction

Four constraints operate independently of code performance.

- **Control and interconnect scaling.** Wiring density, cryogenic cooling capacity and room-temperature control electronics must scale with qubit number. A million-qubit superconducting system cannot be

built by replicating current control architecture; cryogenic CMOS control and photonic interconnects are active research areas precisely because the incumbent approach does not extend.

- **Capital intensity.** Dilution refrigerators, laser systems and ultra-high-vacuum chambers represent multi-million-dollar infrastructure per system, concentrating capability in a small number of well-funded institutions and reinforcing cloud access as the dominant consumption model.
- **Human capital.** Quantum job postings tripled as a share of total postings between 2011 and mid-2024 while the supply of trained researchers grew far more slowly [164]. The interdisciplinary demand — physics, computer science, electrical engineering and materials science in combination — compounds the shortage, since few curricula produce it and the training pipeline is measured in years [165].
- **The utility gap.** Benchmark studies continue to show that many optimisation problems presented as quantum use cases are solvable classically in seconds [10], and scaling to the millions of physical qubits required for commercially relevant applications remains a multi-year engineering programme rather than a research question [147].

9. Market Structure, Public Investment and Roadmaps

9.1 Market Estimates and Their Reliability

The global quantum computing market was valued at approximately \$1.4–\$3.5 billion in 2025, with projections that vary by an order of magnitude depending on methodology: one estimate places the 2025 figure at \$3.52 billion growing to \$20.20 billion by 2030 at 41.8% compound annual growth [148]; another projects \$1.39 billion reaching \$17.89 billion by 2034 [149]; a third forecasts growth from \$1.6 billion to \$7.3 billion by 2030 [150]. A widely cited technology monitor projects total revenue of \$28–\$72 billion by 2035, with vendors generating \$650–\$750 million in 2024 and expected to exceed \$1 billion in 2025 [7].

The dispersion is itself the finding. A twofold disagreement about the present market size and a threefold disagreement about the 2030 figure indicate that analysts do not share a definition of what is being counted — hardware sales, cloud consumption, consulting revenue and government research contracts are variously included or excluded. These projections should be treated as expressions of sentiment rather than as measurements, and a further difficulty is that a substantial share of current revenue derives from government research contracts rather than commercial demand, which makes growth partly a function of public budget cycles.

Investment activity is more concretely measurable. Startup funding reached approximately \$2 billion in 2024, a 50% increase over \$1.3 billion in 2023 [7]. A landmark \$1.68 billion initial public offering in June 2026 valued one trapped-ion developer at approximately \$15 billion [151]; a competitor reported FY2025 revenue of \$130 million, exceeding guidance by 20%, against a market capitalisation of \$13.6 billion [152, 153]; and a photonic developer closed a \$1.5 billion round at a \$10.5 billion valuation [154]. One incumbent has committed more than \$10 billion over five years across research, manufacturing and ecosystem development [155]. The ratio of market capitalisation to revenue across the listed pure-plays — in one case approaching a hundredfold — indicates that public valuations are pricing an option on future capability rather than current earnings, and are correspondingly sensitive to milestone news.

9.2 Public Investment

Government commitments have reached levels that reflect a strategic rather than a scientific framing. Reported figures place national investment at approximately \$15 billion in one case [156], \$7.4 billion in another [7], and £2 billion (roughly \$2.67 billion) in a third [157]. Legislative incentive programmes

account for more than \$2 billion in federal support, with letters of intent signed across several domestic manufacturers [158]. Additional national commitments include approximately \$900 million, \$620 million directed to a single manufacturing facility, \$500 million for a regional quantum park, and approximately \$222 million [7]. A regional strategy targets global leadership by 2030 [159].

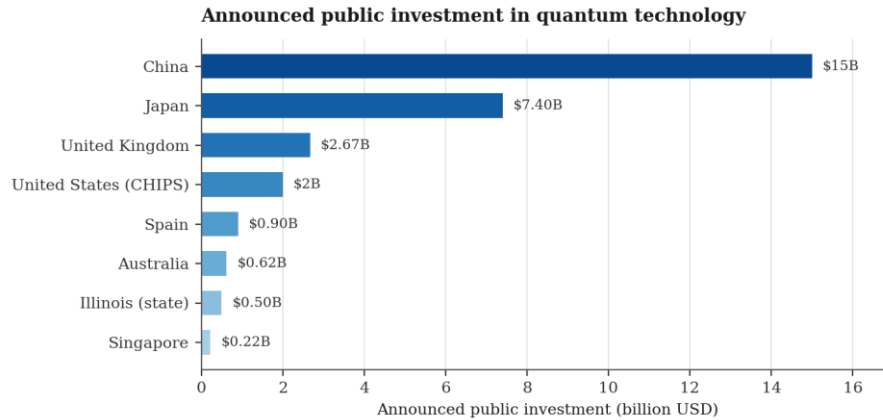


Figure 2. Announced public investment in quantum technology by jurisdiction. Figures are announced commitments across differing time horizons and accounting conventions and are not directly comparable as annual spend.

Two dynamics follow from the scale of public funding. First, national-security framing accelerates funding but also invites export controls and restrictions on international collaboration, which cut against a field that has historically depended on open publication. Second, where public money constitutes a large fraction of demand, apparent market growth may reflect appropriations rather than adoption — a distinction that matters for anyone using market projections to plan.

9.3 Corporate Roadmaps and Enterprise Sentiment

Published roadmaps converge on the end of the decade. One vendor plans a fault-tolerant system executing 100 million gates on 200 logical qubits by 2029, followed by a machine running one billion gates on 2,000 logical qubits by 2033 [160], and has targeted demonstrable quantum advantage by the end of 2026 [161]. Another anticipates a scalable topological machine by 2029 [162], and a third a fully fault-tolerant universal computer by 2030 [143].

Enterprise sentiment has diverged from vendor optimism. Reporting indicates that 46% of enterprises expect flat quantum spending in 2026, described as a shift from fear-of-missing-out procurement toward demand for demonstrable return [163]. This is a healthy correction rather than a retreat: it moves evaluation from narrative to evidence, and it disciplines vendors toward problems with measurable baselines. It also implies that the next two years will be judged on delivered logical-qubit counts rather than on physical ones.

10. Critical Synthesis and Research Agenda

10.1 Reconciling Advantage and Utility

The apparent contradiction between demonstrated quantum advantage and commercial disappointment dissolves once the two claims are stated precisely. The 2026 advantage demonstrations concern problems constructed so that quantum evolution is the natural description and classical simulation is provably or empirically expensive [4, 12]. Commercial optimisation problems have the opposite character: they are

classical in origin, admit decades of specialised heuristic development, and possess exploitable structure that quantum formulations typically discard in the process of encoding [10].

The correct inference is not that commercial value is unattainable but that it will emerge first where the problem is already quantum mechanical — chemistry, materials, and the characterisation of quantum devices themselves — and only later, if at all, in generic optimisation. Portfolio construction and vehicle routing were selected as early targets because they are commercially salient, not because they are quantum-suited, and the negative results in those domains are best read as a correction of target selection rather than as evidence about the technology.

10.2 Open Problems

Five problems most tightly constrain the trajectory.

6. **Barren plateaus.** No general solution exists to the exponential decay of gradient variance in variational circuits, and the several distinct mechanisms producing it — expressibility, entanglement and noise — admit no common remedy [46, 47, 119]. Absent a resolution, the entire variational programme lacks a scaling argument.
7. **Encoding overhead.** Even with an approximate tenfold improvement from qLDPC constructions [137], physical-to-logical ratios remain in the hundreds. Reducing this ratio has larger practical effect than adding physical qubits, and hardware-level error bias offers a complementary route [141].
8. **Absence of NISQ-era algorithms with demonstrated advantage.** No algorithm has been shown to outperform tuned classical heuristics on a real-world problem using pre-fault-tolerant hardware. Whether such an algorithm exists is an open question with substantial commercial consequences.
9. **Verification.** Establishing that a quantum computation is correct when it cannot be classically checked is a foundational problem; the 2026 verifiable-advantage results represent progress but do not constitute a general framework [4].
10. **Workforce.** The talent shortage is structural rather than cyclical, since training a quantum engineer takes years and the required combination of disciplines is produced by few institutions [164, 165].

10.3 Recommendations

For research organisations, the highest-return activity is problem-selection discipline: establishing a rigorous classical baseline before committing to a quantum formulation, since the majority of reported quantum advantages in the applied literature have not survived that test. For enterprises, the defensible near-term posture is to treat cryptographic migration as urgent and operational, and quantum computing itself as a monitoring activity with limited pilot exposure concentrated in chemistry and materials. For policymakers, the binding constraint over the next five years is human capital rather than capital expenditure; funding directed at doctoral and technician training is likely to have greater marginal effect than funding directed at additional hardware.

10.4 Limitations of This Review

Three limitations qualify the conclusions. First, a substantial fraction of the performance figures reported here are vendor-supplied and measured under conditions the vendor selected; no independent cross-platform benchmarking authority currently exists with the standing to audit them. Second, preprint sources have not completed peer review, and results in this field have occasionally been revised downward following scrutiny. Third, publication bias toward positive results is severe, and the small number of negative results cited here — the portfolio benchmark and the resource-estimation-driven programme

closure — are almost certainly not representative of the full population of unsuccessful efforts. Conclusions should be adjusted accordingly, in the pessimistic direction.

11. Conclusion

Between 2024 and 2026 quantum computing acquired something it previously lacked: a body of verifiable results. Surface-code memories were shown to improve as they grow rather than degrade [133]; verifiable computational advantage was demonstrated on problems that resist classical simulation [4, 12]; simulation results were validated against independent physical measurement [115]; and the first post-quantum cryptographic standards were finalised and are entering deployment [6, 108].

In the same interval the field acquired a falsification record. Optimisation problems widely presented as quantum use cases were solved classically in seconds [10]; a major financial institution withdrew after resource estimation placed its target problem beyond any plausible machine [13]; and enterprise budgets flattened as procurement shifted from anticipation to evidence [163].

Both records are accurate, and the tension between them is the defining feature of the present moment. Scientific advantage exists; economic advantage does not yet. The interval between them will be determined by three quantities that are measurable and should be tracked in preference to qubit counts: the physical-to-logical encoding ratio, the logical error rate per cycle, and the number of trained practitioners entering the field annually. Cryptographic migration is the one area where the evidence supports immediate and unambiguous action, and it should not be deferred pending resolution of the broader questions.

The most probable trajectory is neither the collapse implied by the negative results nor the transformation implied by the roadmaps, but a narrower and slower path in which quantum processors become specialised scientific instruments — valuable for simulating quantum systems, largely irrelevant to general-purpose computation — with commercial applications emerging first in chemistry and materials rather than in the optimisation problems that have attracted the most attention and delivered the least.

References

Sources are ordered by first citation. Peer-reviewed literature and standards documents are cited in preference to secondary reporting; where a performance claim originates with a vendor, the source is identified as such.

- [1] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th Anniversary ed., Cambridge University Press, 2010.
- [2] D. P. DiVincenzo, "The physical implementation of quantum computation," *Fortschritte der Physik*, vol. 48, pp. 771–783, 2000.
- [3] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018. arXiv:1801.00862.
- [4] Google, "Quantum Echoes: Willow and verifiable quantum advantage." <https://blog.google/innovation-and-ai/technology/research/quantum-echoes-willow-verifiable-quantum-advantage/>
- [5] IBM Quantum documentation, "Processor types." <https://docs.quantum.ibm.com/guides/processor-types>
- [6] NIST, "Post-Quantum Cryptography Standardization." <https://csrc.nist.gov/Projects/post-quantum-cryptography/Post-Quantum-Cryptography-Standardization>
- [7] McKinsey & Company, "The Year of Quantum: From Concept to Reality." <https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/the-year-of-quantum-from-concept-to-reality-in-2025>
- [8] Quantinuum, "Introducing Helios." <https://www.quantinuum.com/blog/introducing-helios-the-most-accurate-quantum-computer-in-the-world>
- [9] Atom Computing, "Quantum computing technology." <https://atom-computing.com/quantum-computing-technology/>
- [10] Benchmark study of quantum and classical methods on 250 portfolio instances. arXiv:2509.17876.

- [11] IBM Quantum, "Nighthawk product update," January 2026. <https://quantum.cloud.ibm.com/announcements/product-updates/2026-01-05-nighthawk>
- [12] IBM Newsroom, "IBM and Algorithmiq demonstrate quantum advantage," July 2026. <https://newsroom.ibm.com/2026-07-30-ibm-and-algorithmiq-demonstrate-quantum-advantage>
- [13] Reuters/Yahoo Finance, "Goldman Sachs pulls back from quantum research." <https://ca.finance.yahoo.com/news/goldman-pulls-back-8-million-185000531.html>
- [14] C. M. Dawson and M. A. Nielsen, "The Solovay–Kitaev algorithm," arXiv:quant-ph/0505030.
- [15] Emergent Mind, "Grover's search algorithm." <https://www.emergentmind.com/topics/grover-s-search-algorithm>
- [16] Analysis of Grover iteration counts. arXiv:2402.00082.
- [17] C. H. Bennett, E. Bernstein, G. Brassard and U. Vazirani, "Strengths and weaknesses of quantum computing," *SIAM J. Comput.*, vol. 26, pp. 1510–1523, 1997.
- [18] J. Koch et al., "Charge-insensitive qubit design derived from the Cooper pair box," *Phys. Rev. A*, vol. 76, 042319, 2007.
- [19] IBM Quantum Summit 2023 technical disclosures. <https://ibm.com/quantum/summit-2023>
- [20] Google Quantum AI, "Quantum error correction below the surface code threshold," *Nature*, 2024. <https://www.nature.com/articles/s41586-024-08449-y>
- [21] Google Research, "Making quantum error correction work." <https://research.google/blog/making-quantum-error-correction-work/>
- [22] Survey of superconducting qubit coherence times. arXiv:2407.18778.
- [23] Trapped-ion coherence measurements, *Nature Photonics*. <https://www.nature.com/articles/s41566-017-0007-1>
- [24] Extended coherence in decoherence-free subspaces. arXiv:2603.19631.
- [25] Photonics Online, "Xanadu launches first public cloud-deployed computer with quantum computational advantage."
- [26] Tom's Hardware, "Quantum photonics roadmap: Xanadu and PsiQuantum."
- [27] Tom's Hardware, "Neutral-atom quantum computing roadmap."
- [28] The Quantum Insider, "Atom Computing raises \$60 million Series B," January 2022.
- [29] Parallel two-qubit gates on neutral atom arrays. arXiv:2304.05420.
- [30] D-Wave Systems, "Advantage2 4,400-qubit technical whitepaper."
- [31] Microsoft Quantum, "Majorana 2: a scalable quantum processor." <https://quantum.microsoft.com/en-us/insights/blogs/majorana-2-scalable-quantum-processor>
- [32] HotHardware, "Microsoft Majorana quantum breakthrough."
- [33] *Physics* (APS), commentary on Majorana zero-mode evidence, vol. 18, art. 68. <https://physics.aps.org/articles/v18/68>
- [34] A. W. Cross et al., "Validating quantum computers using randomized model circuits," *Phys. Rev. A*, vol. 100, 032328, 2019; and CLOPS benchmark, arXiv:2110.14108.
- [35] Shor's algorithm: modern exposition and resource analysis. arXiv:2509.05010.
- [36] Y. Burton, "Shor's algorithm implementation study," Stanford CS269Q, 2019.
- [37] S. Beauregard, "Circuit for Shor's algorithm using $2n+3$ qubits," arXiv:quant-ph/0205095.
- [38] Analysis of the general number field sieve. arXiv:2503.10008.
- [39] NIST Fifth PQC Standardization Conference, "On the practical cost of Grover for AES key recovery."
- [40] Optimised factoring resource estimation. arXiv:2512.15330.
- [41] Factorisation of a 48-bit integer on superconducting hardware. arXiv:2212.12372.
- [42] Generalised period decomposition on NISQ devices. arXiv:2512.11004.
- [43] Variational quantum eigensolver methodology. arXiv:2407.16107.
- [44] Variational principle and energy bounds in VQE. arXiv:2411.13742.
- [45] VQE demonstration on ethylene. arXiv:2109.02110.
- [46] Barren plateaus in parameterised circuits. arXiv:2501.07433.
- [47] Unified analysis of barren plateau mechanisms. arXiv:2407.17706.
- [48] QAOA formulation and parameter analysis. arXiv:2410.22247.
- [49] Fixed-angle conjecture for QAOA on 3-regular graphs. arXiv:2107.00677.
- [50] IBM Quantum Learning, "Variational quantum algorithms." <https://quantum.cloud.ibm.com/learning/en/courses/utility-scale-quantum-computing/variational-quantum-algorithms>
- [51] QAOA experiments at 55 qubits. arXiv:2403.04296.
- [52] Quantum simulation algorithms review. arXiv:2310.12256.
- [53] Recent advances in Hamiltonian simulation. arXiv:2510.01710.
- [54] Improved Jordan–Wigner transformation cost. arXiv:1403.1539.
- [55] Wannier-basis Hamiltonians with hybrid fermion-to-qubit mapping, *Nature Communications*, 2023. <https://www.nature.com/articles/s41467-023-43479-6>
- [56] A. M. Childs et al., "Theory of Trotter error with commutator scaling," *Phys. Rev. X*, vol. 11, 011020, 2021.

- [57] E. Tang, "A quantum-inspired classical algorithm for recommendation systems," arXiv:1807.04271.
- [58] IBM Quantum, "Qiskit 1.0 is here," February 2024. <https://docs.quantum.ibm.com/announcements/product-updates/2024-02-15-qiskit-1.0-is-here>
- [59] IBM Quantum, "Qiskit 1.0 release summary." <https://www.ibm.com/quantum/blog/qiskit-1-0-release-summary>
- [60] IBM Quantum for startups — processor availability. <https://www.ibm.com/quantum/startups>
- [61] Google Quantum AI, "Cirq: introduction." <https://quantumai.google/cirq/start/intro>
- [62] Google Quantum AI, "Cirq concepts." <https://quantumai.google/cirq/google/concepts>
- [63] Google Quantum AI, Cirq simulator documentation. <https://quantumai.google/cirq>
- [64] Google Quantum AI, "Cirq devices." <https://quantumai.google/cirq/google/devices>
- [65] Microsoft Learn, "Resource estimation results." <https://learn.microsoft.com/en-us/azure/quantum/qre-estimation-results>
- [66] Microsoft Q# blog, "Design fault-tolerant applications with the open-source Resource Estimator."
- [67] Microsoft Learn, "Quickstart: Microsoft Resource Estimator." <https://learn.microsoft.com/en-us/azure/quantum/quickstart-microsoft-resources-estimator>
- [68] PennyLane, "Architecture guide." <https://docs.pennylane.ai/en/latest/development/guide/architecture.html>
- [69] PennyLane documentation index. <https://docs.pennylane.ai/en/latest/index.html>
- [70] Xanadu, "PennyLane." <https://www.xanadu.ai/pennylane>
- [71] QuTiP source repository. <https://github.com/qutip>
- [72] QuTiP 5.0 documentation. <https://qutip.readthedocs.io/en/latest/>
- [73] IBM Quantum, "Plans overview." <https://docs.quantum.ibm.com/guides/plans-overview>
- [74] IBM Quantum, "Utility-scale quantum credits." <https://www.ibm.com/quantum/blog/utility-scale-quantum-credits>
- [75] IBM Quantum access plans. <https://www.ibm.com/quantum/access-plans>
- [76] AWS, "Amazon Braket regions." <https://docs.aws.amazon.com/braket/latest/developerguide/braket-regions.html>
- [77] AWS, "Amazon Braket pricing." <https://aws.amazon.com/braket/pricing/>
- [78] AWS Quantum Computing Blog, "Amazon Braket launches Rigetti Cepheus-1 108Q superconducting device."
- [79] Microsoft Learn, "Quantinuum provider." <https://learn.microsoft.com/en-us/azure/quantum/provider-quantinuum>
- [80] Microsoft Learn, "IonQ provider." <https://learn.microsoft.com/en-us/azure/quantum/provider-ionq>
- [81] Microsoft Learn, "Azure Quantum credits." <https://learn.microsoft.com/azure/quantum/azure-quantum-credits>
- [82] Google Quantum AI, "Willow Early Access Program." <https://quantumai.google/willowearlyaccess>
- [83] Google Quantum AI, "Getting started with Cirq on Google hardware." <https://quantumai.google/cirq/tutorials/google/start>
- [84] Review of quantum key distribution, *Nature Reviews Electrical Engineering*. <https://www.nature.com/articles/s44287-025-00238-7>
- [85] Tech Times, "Quantum key distribution spans 303 km over live Swedish fibre," June 2026.
- [86] Entanglement-based QKD security analysis, *Springer*. <https://link.springer.com/article/10.1007/s10791-025-09807-8>
- [87] ETSI/IQC Quantum-Safe Workshop, technical track on QKD networks, 2018.
- [88] Large-scale integrated QKD network, *npj Quantum Information*. <https://www.nature.com/articles/s41534-025-01089-8>
- [89] Phys.org, "World-first quantum microsatellite communication with multiple ground stations," March 2025.
- [90] European Commission, "European Quantum Communication Infrastructure (EuroQCI)." <https://digital-strategy.ec.europa.eu/en/policies/european-quantum-communication-infrastructure-euroqci/>
- [91] JPMorgan Chase Technology Blog, "Building a QKD network for a critical blockchain application."
- [92] NIST, FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard. <https://csrc.nist.gov/pubs/fips/203/final>
- [93] NIST, FIPS 204: Module-Lattice-Based Digital Signature Standard. <https://csrc.nist.gov/pubs/fips/204/final>
- [94] NIST, FIPS 205: Stateless Hash-Based Digital Signature Standard. <https://www.nist.gov/publications/stateless-hash-based-digital-signature-standard>
- [95] Encryption Consulting, "FN-DSA (FIPS 206) overview." <https://www.encryptionconsulting.com/education-center/fn-dsa-fips-206/>
- [96] NIST, "NIST selects HQC as fifth algorithm for post-quantum encryption," March 2025.
- [97] Performance evaluation of ML-KEM on constrained hardware. arXiv:2603.19340.
- [98] The Quantum Insider, "Q-Day just got closer: three papers rewriting the quantum threat timeline," March 2026.
- [99] Global Risk Institute / Quantum Security Defence, "Q-Day probability research synthesis."
- [100] Quantum Computing Report, "Q-Day accelerated timeline across a wider attack surface."
- [101] TechRadar, survey reporting on harvest-now-decrypt-later concern. <https://www.techradar.com>
- [102] NIST IR 8547 (initial public draft), "Transition to post-quantum cryptography standards." <https://csrc.nist.gov/pubs/ir/8547/ipd>
- [103] The Quantum Insider, "Post-quantum migration timelines: government and industry impact," May 2026.
- [104] Encryption Consulting, "What is CNSA 2.0?" <https://www.encryptionconsulting.com/education-center/what-is-cnsa-2-0/>

- [105] PostQuantum.com, "US PQC regulatory framework 2026." <https://postquantum.com/quantum-policies/us-pqc-regulatory-framework-2026/>
- [106] ETSI TR 104 016, "Quantum-safe cryptography: migration framework."
- [107] ETSI TS 103 744, "Quantum-safe hybrid key establishment."
- [108] Cloudflare Research, "The state of the post-quantum Internet." <https://blog.cloudflare.com/pq-2024/>
- [109] IBM Newsroom, "Cleveland Clinic, RIKEN and IBM model a 12,635-atom protein," May 2026.
- [110] The Quantum Insider, "PsiQuantum and Boehringer Ingelheim report speedups for complex molecular simulations," January 2025.
- [111] IBM, "Moderna case study." <https://www.ibm.com/case-studies/moderna>
- [112] Boehringer Ingelheim Annual Report 2021, quantum computing feature.
- [113] Investing.com, "JPMorgan and Amazon develop quantum tools for finance problems."
- [114] The Quantum Insider, "Hybrid quantum algorithm improves portfolio optimisation on trapped-ion hardware," July 2026.
- [115] IBM Newsroom, "IBM quantum computer accurately simulates real magnetic materials," March 2026.
- [116] Quantum embedding for electrode surface reactions. arXiv:2503.10923.
- [117] Quantum modelling of strongly correlated transition-metal oxide catalysts. arXiv:2512.19778.
- [118] Hybrid solvers for capacitated vehicle routing with time windows. arXiv:2503.24285.
- [119] Trainability tension between deep and shallow variational circuits. arXiv:2405.05332.
- [120] Mitigation strategies for barren plateaus. arXiv:2406.14285.
- [121] Quantum kernel methods in financial applications. arXiv:2405.10119.
- [122] JPMorgan Chase Technology Blog, "OQC, JPMC and AMD quantum AI platform."
- [123] H.-Y. Huang et al., "Power of data in quantum machine learning," *Nature Communications*, vol. 12, 2631, 2021.
- [124] Google Research, "Making quantum error correction work" (extended technical discussion).
- [125] Crosstalk-induced phase transitions in surface-code architectures, *npj Quantum Information*. <https://www.nature.com/articles/s41534-026-01241-y>
- [126] IBM Quantum documentation, "Error mitigation and suppression techniques."
- [127] Bivariate bicycle qLDPC codes, *Nature*, 2024. <https://www.nature.com/articles/s41586-024-07107-7>
- [128] Logical error-rate requirements for useful algorithms, *Nature Communications*. <https://www.nature.com/articles/s41467-025-59714-1>
- [129] Exponential suppression of bit or phase errors with repetition codes, *Nature*, 2021. <https://www.nature.com/articles/s41586-021-03588-y>
- [130] Realizing repeated quantum error correction in a distance-3 surface code, *Nature*, 2022. <https://www.nature.com/articles/s41586-022-04566-8>
- [131] Logical quantum processor based on reconfigurable atom arrays, *Nature*, 2023. <https://www.nature.com/articles/s41586-023-06927-3>
- [132] Microsoft Quantum, "Application of error correction to trapped-ion qubits."
- [133] Semiconductor Engineering, "Google's surface code technique to reduce errors."
- [134] Universal fault-tolerant architecture on 448 neutral atoms, *Nature*, 2025. <https://www.nature.com/articles/s41586-025-09848-5>
- [135] Microsoft Azure Quantum, "How Microsoft and Quantinuum achieved reliable quantum computing," April 2024.
- [136] The Quantum Insider, "QuEra, Harvard and MIT demonstrate logical-level magic state distillation," July 2025.
- [137] IBM Quantum, "Nature qLDPC error correction." <https://www.ibm.com/quantum/blog/nature-qldpc-error-correction>
- [138] Google Research, "A colorful quantum future." <https://research.google/blog/a-colorful-quantum-future/>
- [139] Quantinuum Tesseract code and fault-tolerant logical operations. arXiv:2409.04628.
- [140] D. Aharonov and M. Ben-Or, "Fault-tolerant quantum computation with constant error rate," arXiv:quant-ph/9906129.
- [141] H. Putterman et al., "Hardware-efficient quantum error correction using concatenated bosonic qubits," *Nature*, vol. 638, 2025.
- [142] IBM Quantum, "Large-scale fault-tolerant quantum computing." <https://www.ibm.com/quantum/blog/large-scale-ftqc>
- [143] Quantinuum, "By the end of the decade we will deliver universal fault-tolerant quantum computing."
- [144] IBM Quantum documentation, "Configure error mitigation."
- [145] IBM Research, "Recent experimental progress on quantum error mitigation for large quantum circuits."
- [146] IBM Research Blog, "Eagle and quantum error mitigation."
- [147] The Quantum Insider, "Understanding quantum error correction," March 2026.
- [148] MarketsandMarkets, "Quantum computing market report." <https://www.marketsandmarkets.com/Market-Reports/quantum-computing-market-144888301.html>
- [149] Fortune Business Insights, "Quantum computing market." <https://www.fortunebusinessinsights.com/quantum-computing-market-104855>

- [150] BCC Research, "Global quantum computing market to grow."
- [151] Yahoo Finance, "Quantinuum IPO and sector valuation movements," June 2026.
- [152] IonQ Investor Relations, "Fourth quarter and full year 2025 financial results."
- [153] Investing.com, "IonQ vs Rigetti: which quantum computing stock has more upside."
- [154] Sacra, "PsiQuantum company profile and funding history." <https://sacra.com/c/psiquantum/>
- [155] IBM Newsroom, "IBM commits more than \$10 billion to quantum computing," June 2026.
- [156] Broadband Breakfast, "As federal quantum funding lapses, states deploy quantum-secured fibre."
- [157] Quantum Computing Report, "UK government commits £2 billion to national quantum scaling and procurement."
- [158] JD Supra, "GovCon update: quantum funding." <https://www.jdsupra.com/legalnews/govcon-update-quantum-funding-fixed-4994002/>
- [159] Infosecurity Magazine, "EU plan for a quantum-secure Europe."
- [160] IBM Quantum, "\$10 billion investment FAQ." <https://www.ibm.com/quantum/blog/10-billion-investment-faq>
- [161] IBM Newsroom, "IBM delivers new quantum processors, software and algorithm breakthroughs," November 2025.
- [162] Microsoft Source, "Majorana 2 and Microsoft discovery," 2025.
- [163] Quantum Computing Report, "Quantum usage entering a 'show me' phase as executives demand measurable ROI."
- [164] MIT Sloan, "Building a quantum workforce." <https://mitsloan.mit.edu/ideas-made-to-matter/building-a-quantum-workforce>
- [165] EPJ Quantum Technology, "Quantum workforce development and interdisciplinary training," 2024. <https://link.springer.com/10.1140/epjqt/s40507-024-00294-2>